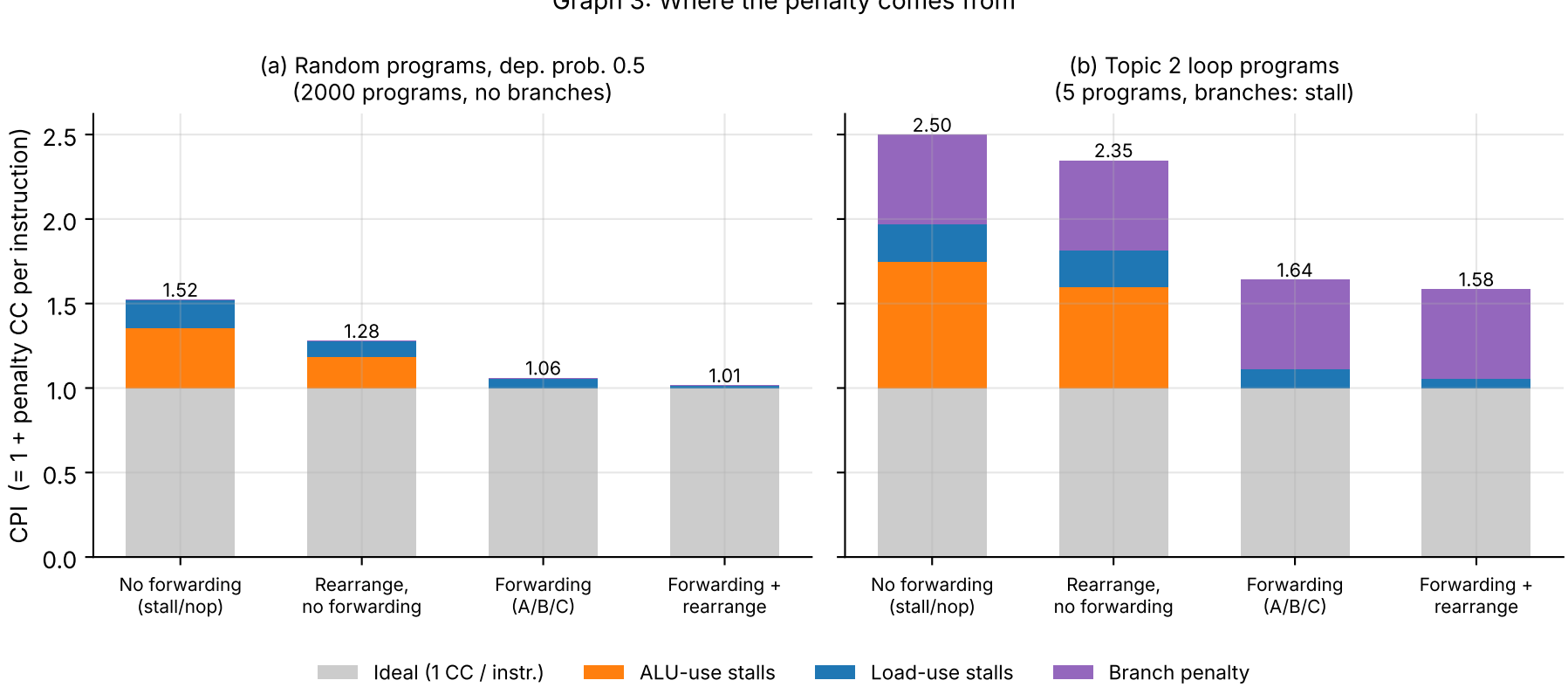
Graph 3: Where the penalty comes from
(a) Random programs, dep. prob. 0.5
(2000 programs, no branches)
(b) Topic 2 loop programs
(5 programs, branches: stall)
CPI (= 1 + penalty CC per instruction)
2.5
2.0
1.5
1.0
0.5
0.0
1.52
1.28
1.06
1.01
2.50
2.35
1.64
1.58
No forwarding (stall/nop)
Rearrange, no forwarding
Forwarding (A/B/C)
Forwarding + rearrange
Ideal (1 CC / instr.)
ALU-use stalls
Load-use stalls
Branch penalty